

**Phase 3 FortiGate Hybrid WAN
Architecture: MPLS-BGP (Primary) &
IPsec VPN (Backup)**

Engr. Sam Laurence Manapsal ECT,CCNA

FortiGate Hybrid WAN Architecture: MPLS-BGP (Primary) & IPsec VPN (Backup)

This project simulates a real-world enterprise WAN environment designed to provide secure, reliable, and redundant communication between branch networks. The network uses MPLS as the primary WAN connection for fast and private branch-to-branch communication. Two ISP connections are also implemented to simulate internet connectivity and provide backup transport during WAN failures. To secure communication over the public internet, a Site-to-Site IPsec VPN tunnel is configured between FortiGate firewalls. This allows encrypted traffic to continue flowing even if the MPLS connection becomes unavailable.

What it is

The network simulates an enterprise environment where two customer branches communicate securely using:

- MPLS WAN connectivity
- ISP internet connectivity
- IPsec VPN tunnel
- Dynamic routing protocols

The topology includes:

- Customer Edge Routers (CE)
- Provider Edge Routers (PE)
- Provider Core Router (P)
- ISP Routers
- FortiGate Firewalls

Large enterprise networks require:

- Secure branch connectivity
- WAN redundancy
- Encrypted internet communication
- Dynamic routing scalability
- Automatic failover mechanisms

This lab simulates how enterprise organizations maintain branch communication even during WAN outages.

Why it exists

This architecture exists because it combines the flawless day-to-day performance of private circuits with the cost-effective reliability of the public internet for emergency backup. During normal operations, the firewall routes your critical corporate data over the premium MPLS path to take full advantage of its guaranteed speeds, zero packet loss, and rock-solid stability. If that primary line physically fails, the firewall immediately shifts traffic to the cheaper ISP link, instantly spinning up an encrypted Site-to-Site IPsec VPN to securely protect your corporate data across the public web. This automated handoff ensures your business stays completely online and fully secure without requiring expensive, redundant private links at every location.

How it works together

Primary Path

- Traffic uses the MPLS provider network as the preferred WAN transport.
- FortiGate → CE → PE → P → PE → CE → FortiGate

Secondary Path

- If MPLS becomes unavailable, traffic automatically uses the ISP internet connection through an encrypted IPsec VPN tunnel.
- FortiGate → ISP-1 → INTERNET → ISP-2 → VPN Tunnel → FortiGate

Routing Behavior

- OSPF handles internal route advertisement
- MP-BGP exchanges customer routes between PE routers
- Static routes prioritize MPLS
- Floating/static backup routes activate VPN failover
- IPsec encrypts internet traffic between branches

Configurations

FortiGate UI Configurations:

FortiGate Network Interfaces

FortiGate VM64-KVM Create NewEditDeleteIntegrate InterfaceSearchGroup By Type <table><tr><th>Name</th><th>Type</th><th>Members</th><th>IP/Netmask</th></tr><tr><td colspan="4">802.3ad Aggregate</td></tr><tr><td>fortilink</td><td>802.3ad Aggregate</td><td></td><td>Dedicated to FortiSwitch</td></tr><tr><td colspan="4">Physical Interface</td></tr><tr><td>BRANCH-LAN (port3)</td><td>Physical Interface</td><td></td><td>172.16.2.1/255.255.255.0</td></tr><tr><td>ISP-INTERNET (port1)</td><td>Physical Interface</td><td></td><td>16.1.1.1/255.255.255.252</td></tr><tr><td>MGMT-FORTIGATE (port4)</td><td>Physical Interface</td><td></td><td>192.168.58.133/255.255.255.0</td></tr><tr><td>MPLS (port2)</td><td>Physical Interface</td><td></td><td>192.168.4.2/255.255.255.0</td></tr><tr><td colspan="4">Tunnel Interface</td></tr><tr><td>NAT Interface (natroot)</td><td>Tunnel Interface</td><td></td><td>0.0.0.0/0.0.0.0</td></tr></table> Security Rating IssuesUpdated: 00:06:49	Name	Type	Members	IP/Netmask	802.3ad Aggregate				fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	Physical Interface				BRANCH-LAN (port3)	Physical Interface		172.16.2.1/255.255.255.0	ISP-INTERNET (port1)	Physical Interface		16.1.1.1/255.255.255.252	MGMT-FORTIGATE (port4)	Physical Interface		192.168.58.133/255.255.255.0	MPLS (port2)	Physical Interface		192.168.4.2/255.255.255.0	Tunnel Interface				NAT Interface (natroot)	Tunnel Interface		0.0.0.0/0.0.0.0	FortiGate VM64-KVM Create NewEditDeleteIntegrate InterfaceSearchGroup By Type <table><tr><th>Name</th><th>Type</th><th>Members</th><th>IP/Netmask</th></tr><tr><td colspan="4">802.3ad Aggregate</td></tr><tr><td>fortilink</td><td>802.3ad Aggregate</td><td></td><td>Dedicated to FortiSwitch</td></tr><tr><td colspan="4">Physical Interface</td></tr><tr><td>HQ-LAN (port3)</td><td>Physical Interface</td><td></td><td>172.16.1.1/255.255.255.0</td></tr><tr><td>ISP-INTERNET (port1)</td><td>Physical Interface</td><td></td><td>15.1.1.1/255.255.255.252</td></tr><tr><td>MGMT-FORTIGATE (port4)</td><td>Physical Interface</td><td></td><td>192.168.58.132/255.255.255.0</td></tr><tr><td>MPLS (port2)</td><td>Physical Interface</td><td></td><td>192.168.3.2/255.255.255.0</td></tr><tr><td colspan="4">Tunnel Interface</td></tr><tr><td>NAT Interface (natroot)</td><td>Tunnel Interface</td><td></td><td>0.0.0.0/0.0.0.0</td></tr></table> Security Rating IssuesUpdated: 00:06:58	Name	Type	Members	IP/Netmask	802.3ad Aggregate				fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	Physical Interface				HQ-LAN (port3)	Physical Interface		172.16.1.1/255.255.255.0	ISP-INTERNET (port1)	Physical Interface		15.1.1.1/255.255.255.252	MGMT-FORTIGATE (port4)	Physical Interface		192.168.58.132/255.255.255.0	MPLS (port2)	Physical Interface		192.168.3.2/255.255.255.0	Tunnel Interface				NAT Interface (natroot)	Tunnel Interface		0.0.0.0/0.0.0.0
Name	Type	Members	IP/Netmask																																																																														
802.3ad Aggregate																																																																																	
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch																																																																														
Physical Interface																																																																																	
BRANCH-LAN (port3)	Physical Interface		172.16.2.1/255.255.255.0																																																																														
ISP-INTERNET (port1)	Physical Interface		16.1.1.1/255.255.255.252																																																																														
MGMT-FORTIGATE (port4)	Physical Interface		192.168.58.133/255.255.255.0																																																																														
MPLS (port2)	Physical Interface		192.168.4.2/255.255.255.0																																																																														
Tunnel Interface																																																																																	
NAT Interface (natroot)	Tunnel Interface		0.0.0.0/0.0.0.0																																																																														
Name	Type	Members	IP/Netmask																																																																														
802.3ad Aggregate																																																																																	
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch																																																																														
Physical Interface																																																																																	
HQ-LAN (port3)	Physical Interface		172.16.1.1/255.255.255.0																																																																														
ISP-INTERNET (port1)	Physical Interface		15.1.1.1/255.255.255.252																																																																														
MGMT-FORTIGATE (port4)	Physical Interface		192.168.58.132/255.255.255.0																																																																														
MPLS (port2)	Physical Interface		192.168.3.2/255.255.255.0																																																																														
Tunnel Interface																																																																																	
NAT Interface (natroot)	Tunnel Interface		0.0.0.0/0.0.0.0																																																																														

FortiGate Static Routes

The image displays two side-by-side screenshots of the FortiGate VM64-KVM web interface, specifically the Static Routes configuration page. Both screenshots show a browser window with the URL 192.168.58.133/ng/routin... and a top navigation bar with options like Dashboard, Network, Interfaces, DNS, Packet Capture, SD-WAN, Static Routes, Policy Routes, RIP, OSPF, BGP, Routing Objects, Multicast, Policy & Objects, Security Profiles, VPN, User & Authentication, System, Security Fabric, and Log & Report.

Left Screenshot (Branch HQ Configuration):

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0	16.1.1.2	ISP-INTERNET (port1)	Enabled	
172.16.4.0/24	15.1.1.1	IPSEC BRANCH-HQ	Enabled	
172.16.1.0/24	15.1.1.1	IPSEC BRANCH-HQ	Enabled	

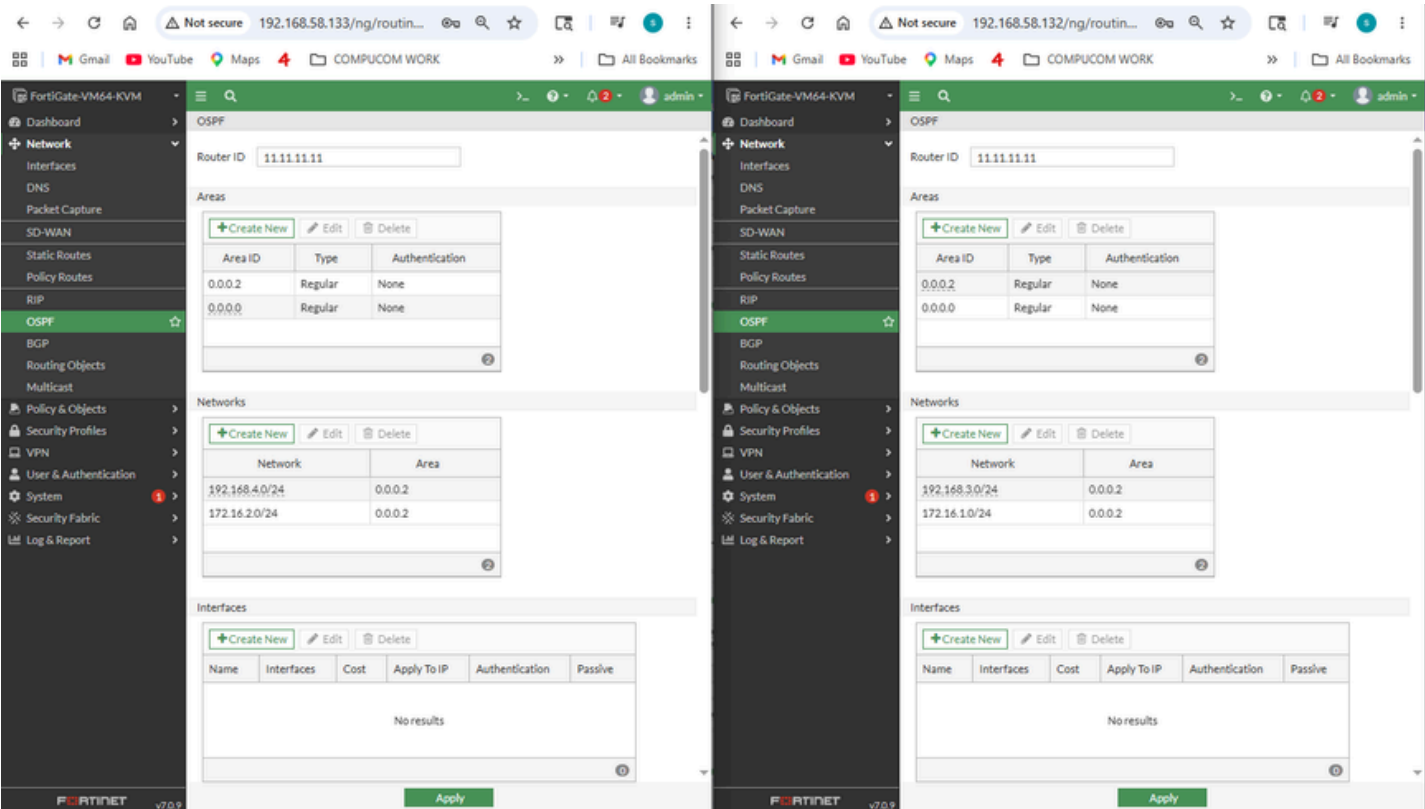
Right Screenshot (Headquarters Configuration):

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0	15.1.1.2	ISP-INTERNET (port1)	Enabled	
172.16.3.0/24	16.1.1.1	IPSEC HQ-BRANCH	Enabled	
172.16.2.0/24	16.1.1.1	IPSEC HQ-BRANCH	Enabled	

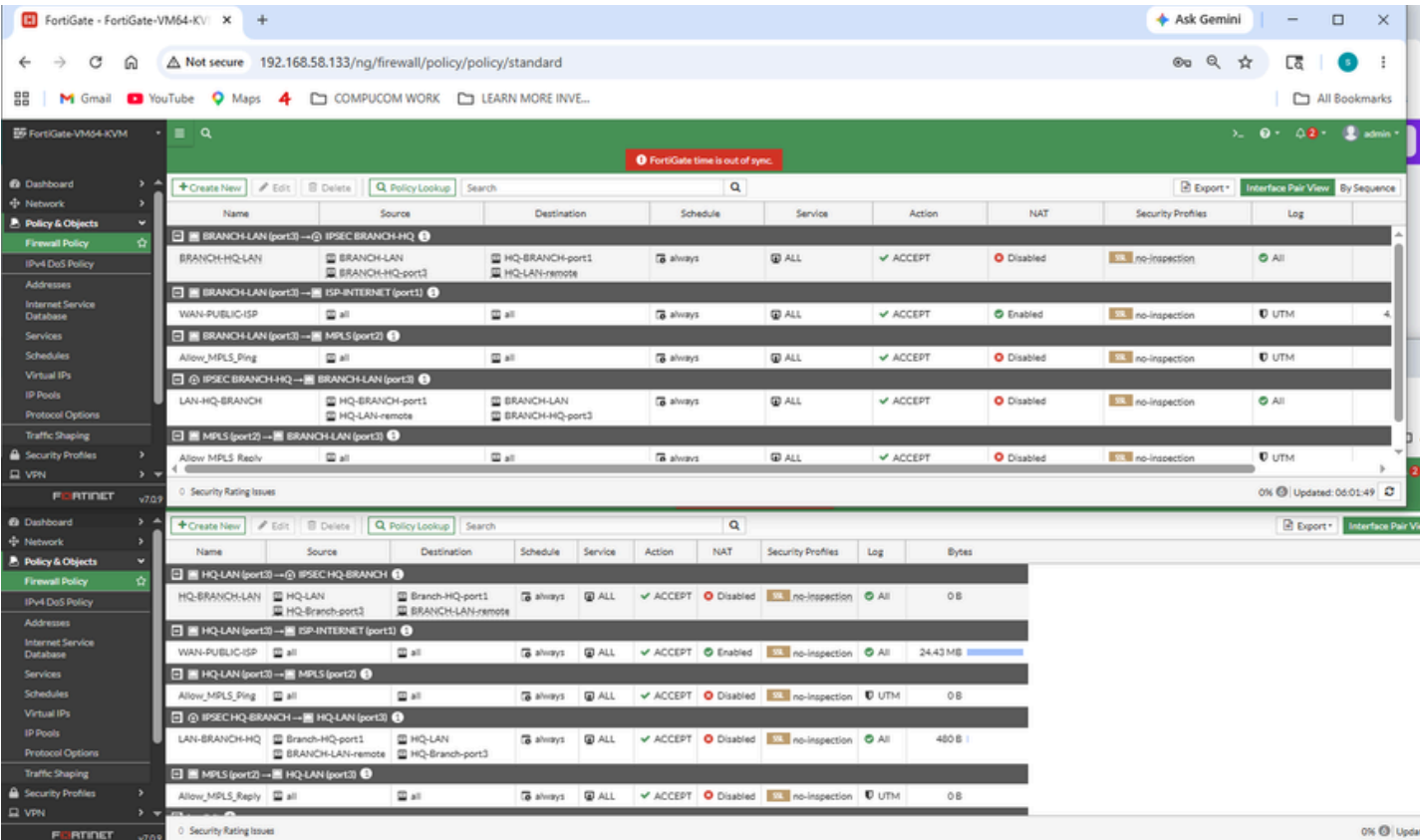
Configurations

FortiGate UI Configurations:

FortiGate OSPF Routes



FortiGate Firewall Policy



Configurations

FortiGate UI Configurations:

FortiGate Addresses

FortiGate-VM64-KVM

Not secure 192.168.58.133/ng/firewal...

FortiGate-VM64-KVM

Not secure 192.168.58.132/ng/firewal...

Name	Details	Interface	Type	Ref
IP Range/Subnet				
BRANCH-HQ-port3	172.16.2.0/24		Address	2
BRANCH-LAN	172.16.3.0/24		Address	2
FABRIC_DEVICE	0.0.0.0/0		Address	0
FIREWALL_AUTH_PORTAL_A...	0.0.0.0/0		Address	0
HQ-BRANCH-port1	172.16.1.0/24		Address	2
HQ-LAN-remote	172.16.4.0/24		Address	2
SSLVPN_TUNNEL_ADDR1	10.212.134.200 - 10.212.134.210		Address	1
all	0.0.0.0/0		Address	4
none	0.0.0.0/32		Address	0

IPsec

Address Group

G Suite

gmail.com
wildcard.google.com

Address Group

0

Microsoft Office 365

login.microsoftonline.com
login.microsoft.com
login.windows.net

Address Group

0

IPsec

Address Group

G Suite

gmail.com
wildcard.google.com

Address Group

0

IPSEC HQ-Branch_remote

Branch-HQ-port1

Address Group

0

Microsoft Office 365

login.microsoftonline.com
login.microsoft.com
login.windows.net

Address Group

0

FortiGate IPsec Tunnel

FortiGate - FortiGate-VM64-KV...

Not secure 192.168.58.133/ng/vpn/ip...

FortiGate - FortiGate-VM64-KV...

Not secure 192.168.58.132/ng/vpn/ip...

Tunnel	Interface Binding	Status	Ref
Custom			
IPSEC BRANCH-HQ	ISP-INTERNET (port1)	Up	0

VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

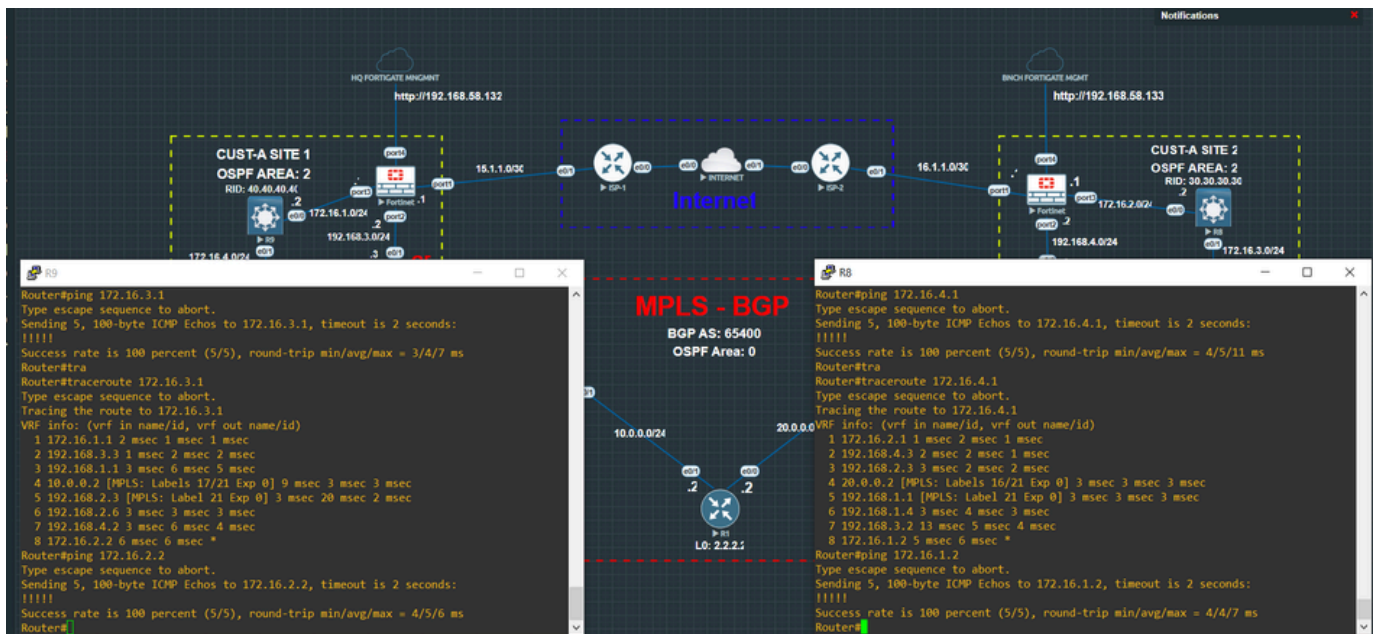
Log & Report

Result

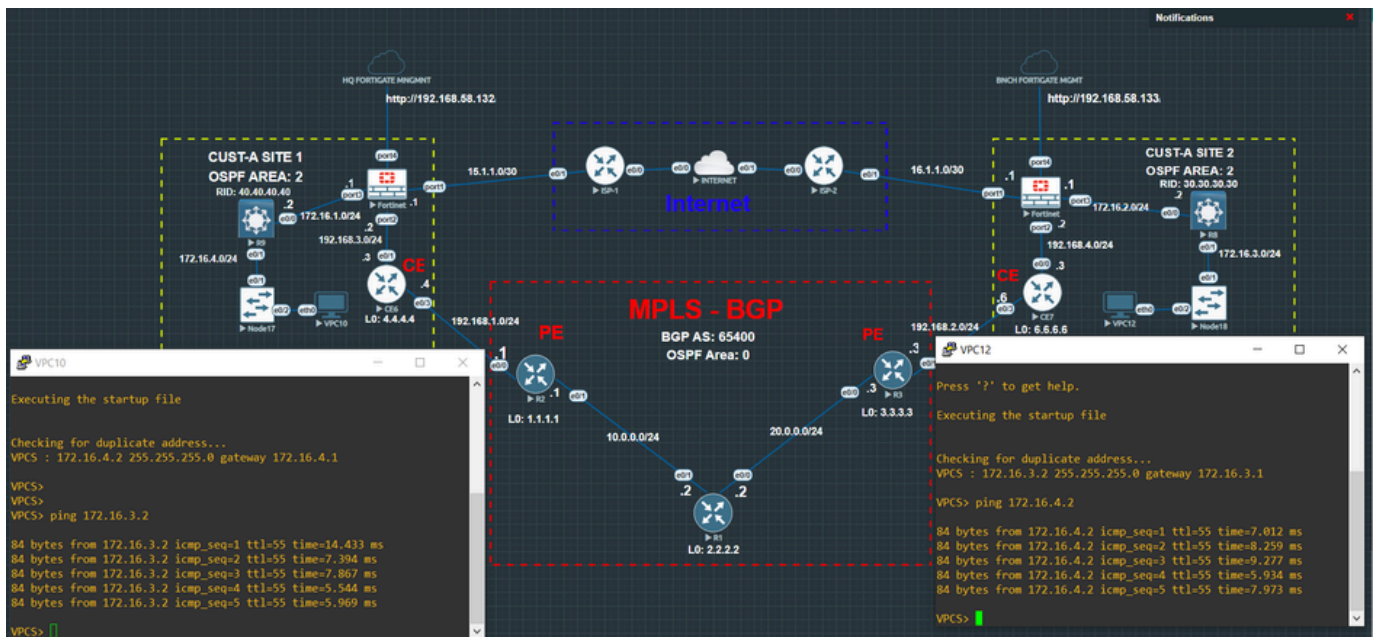
Successfully Implemented

- MPLS WAN Architecture
- OSPF Multi-Area Routing
- MP-BGP Route Exchange
- FortiGate Integration
- Site-to-Site VPN
- Dual WAN Connectivity
- Automatic Failover Logic
- Primary and Secondary WAN Design
- End-to-End Encrypted Communication
- Enterprise WAN Redundancy

PING TEST & TRACEROUTE PASSES ON MPLS PRIMARY



PC-PC PING THROUGH MPLS ROUTE



Result

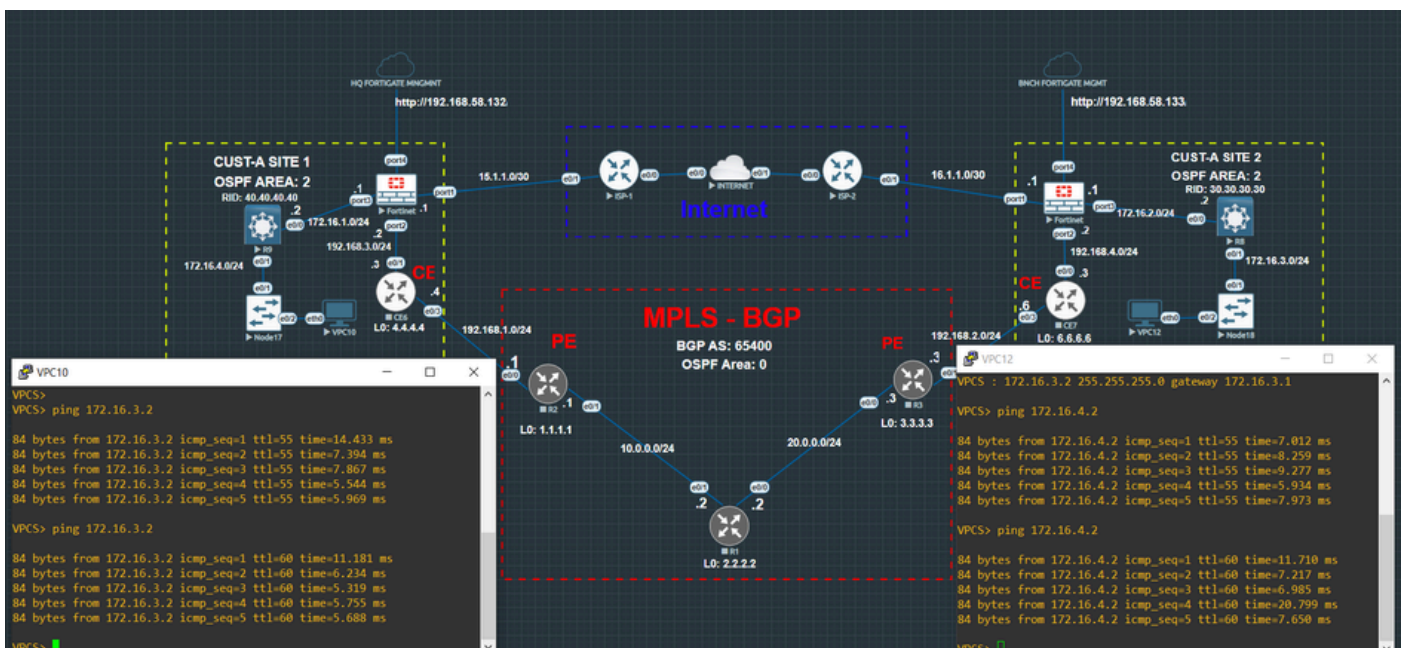
Successfully Implemented

- MPLS WAN Architecture
- OSPF Multi-Area Routing
- MP-BGP Route Exchange
- FortiGate Integration
- Site-to-Site VPN
- Dual WAN Connectivity
- Automatic Failover Logic
- Primary and Secondary WAN Design
- End-to-End Encrypted Communication
- Enterprise WAN Redundancy

PING TEST & TRACEROUTE PASSES ON IPSEC TUNNEL



PC-PC PING THROUGH IPSEC TUNNEL ROUTE



Result

ROUTING TABLE FOR TROUBLESHOOTING

```
Fortinet
FortiGate-VN64-KVM login: admin
Password:
Welcome!

FortiGate-VN64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, LI - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default

Routing table for VRF=0
S* 0.0.0.0/0 [10/0] via 16.1.1.2, port1, [1/0]
O IA 4.4.4.4/32 [110/22] via 192.168.4.3, port2, 02:04:29
O 6.6.6.6/32 [110/21] via 192.168.4.3, port2, 02:04:55
C 16.1.1.0/30 is directly connected, port1
O IA 172.16.1.0/24 [110/32] via 192.168.4.3, port2, 02:04:29
C 172.16.2.0/24 is directly connected, port3
O 172.16.3.0/24 [110/11] via 172.16.2.2, port3, 02:04:55
O IA 172.16.4.0/24 [110/42] via 192.168.4.3, port2, 02:04:29
O IA 192.168.1.0/24 [110/12] via 192.168.4.3, port2, 02:04:29
O 192.168.2.0/24 [110/11] via 192.168.4.3, port2, 02:04:55
O IA 192.168.3.0/24 [110/31] via 192.168.4.3, port2, 02:04:29
C 192.168.4.0/24 is directly connected, port2
C 192.168.58.0/24 is directly connected, port4

FortiGate-VN64-KVM #
```

FW ROUTING TABLE MPLS ACTIVE

```
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
* - replicated route, X - next hop override

Gateway of last resort is not set

4.0.0.0/32 is subnetted, 1 subnets
O 4.4.4.4 [110/12] via 172.16.1.1, 02:03:37, Ethernet0/0
O 6.0.0.0/32 is subnetted, 1 subnets
O IA 6.6.6.6 [110/32] via 172.16.1.1, 02:03:15, Ethernet0/0
172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C 172.16.1.0/24 is directly connected, Ethernet0/0
O IA 172.16.2.0/24 [110/42] via 172.16.1.1, 02:03:15, Ethernet0/0
C 172.16.3.0/24 [110/52] via 172.16.1.1, 02:03:15, Ethernet0/0
C 172.16.4.0/24 is directly connected, Ethernet0/1
L 172.16.4.1/32 is directly connected, Ethernet0/1
O 192.168.1.0/24 [110/21] via 172.16.1.1, 02:03:37, Ethernet0/0
O IA 192.168.2.0/24 [110/22] via 172.16.1.1, 02:03:15, Ethernet0/0
O 192.168.3.0/24 [110/11] via 172.16.1.1, 02:03:37, Ethernet0/0
O IA 192.168.4.0/24 [110/41] via 172.16.1.1, 02:03:15, Ethernet0/0

Router#
```

LAN ROUTERS ROUTING TABLE MPLS ACTIVE

```
Fortinet
FortiGate-VN64-KVM login: admin
Password:
Welcome!

FortiGate-VN64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, LI - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default

Routing table for VRF=0
S* 0.0.0.0/0 [10/0] via 16.1.1.2, port1, [1/0]
C 16.1.1.0/30 is directly connected, port1
C 172.16.1.0/24 [200/0] via IPSEC BRANCH-HQ tunnel 15.1.1.1, [1/0]
C 172.16.2.0/24 is directly connected, port3
S 172.16.2.0/24 [200/0] via IPSEC HQ-BRANCH tunnel 16.1.1.1, [1/0]
S 172.16.3.0/24 [200/0] via IPSEC HQ-BRANCH tunnel 16.1.1.1, [1/0]
O 172.4.0/24 [110/11] via 172.16.1.2, port3, 02:15:07
C 192.168.3.0/24 is directly connected, port2
C 192.168.58.0/24 is directly connected, port4

FortiGate-VN64-KVM # Timeout

FortiGate-VN64-KVM login: 
```

FW ROUTING TABLE IPSEC TUNNEL ACTIVE

```
Router#
Router#
Router#
Router#
Router#
Router#
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, LI - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
* - replicated route, X - next hop override

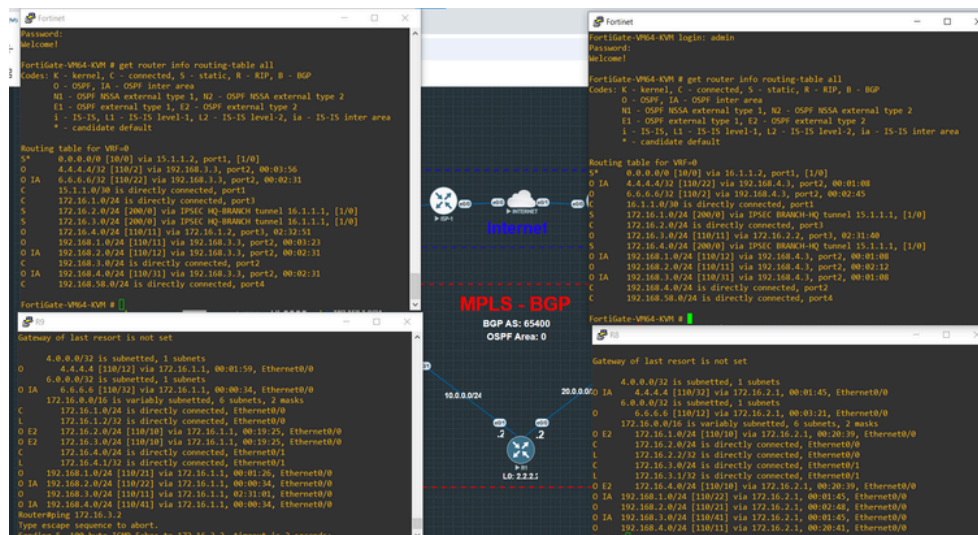
Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C 172.16.1.0/24 is directly connected, Ethernet0/0
L 172.16.1.2/32 is directly connected, Ethernet0/0
O E2 172.16.2.0/24 [110/18] via 172.16.1.1, 00:09:08, Ethernet0/0
O E2 172.16.3.0/24 [110/18] via 172.16.1.1, 00:09:08, Ethernet0/0
L 172.16.4.0/24 is directly connected, Ethernet0/1
L 172.16.4.1/32 is directly connected, Ethernet0/1
O 192.168.3.0/24 [110/11] via 172.16.1.1, 02:20:44, Ethernet0/0

Router#
```

LAN ROUTERS ROUTING TABLE IPSEC TUNNEL ACTIVE

Observations



Even though the primary MPLS circuit recovers and its OSPF routes are successfully re-installed into the routing table, the firewall fails to automatically failback, causing active traffic to remain trapped on the backup IPsec VPN path. This behavior occurs because the FortiGate is a stateful firewall that prioritizes its internal session tracking table; as long as a continuous traffic flow keeps the backup session active, the firewall will not disrupt the socket to re-evaluate the routing table. While deterministic path selection can be manually forced using policy routes to override this hierarchy, the configuration becomes rigid and complex. Without an intelligent SD-WAN framework to actively orchestrate link health monitoring and session preservation, achieving seamless, automated bidirectional failover remains a significant challenge.

Conclusion

Even though the primary MPLS circuit recovers and its OSPF routes are successfully re-installed into the routing table, the firewall fails to automatically failback, causing active traffic to remain trapped on the backup IPsec VPN path. This behavior occurs because the FortiGate is a stateful firewall that prioritizes its internal session tracking table; as long as a continuous traffic flow keeps the backup session active, the firewall will not disrupt the socket to re-evaluate the routing table. While deterministic path selection can be manually forced using policy routes to override this hierarchy, the configuration becomes rigid and complex. Without an intelligent SD-WAN framework to actively orchestrate link health monitoring and session preservation, achieving seamless, automated bidirectional failover remains a significant challenge.

The project demonstrated:

- MPLS provider network deployment
- Dynamic routing using OSPF and MP-BGP
- FortiGate VPN implementation
- WAN redundancy and failover
- Enterprise-grade routing architecture
- Secure encrypted branch communication

The lab significantly improved practical knowledge in:

- Enterprise WAN technologies
- MPLS and BGP operations
- VPN deployment
- Routing troubleshooting
- Network redundancy design
- FortiGate administration
- Service provider architecture

Note

- Same Configurations apply to the other branch just replace the Subnets